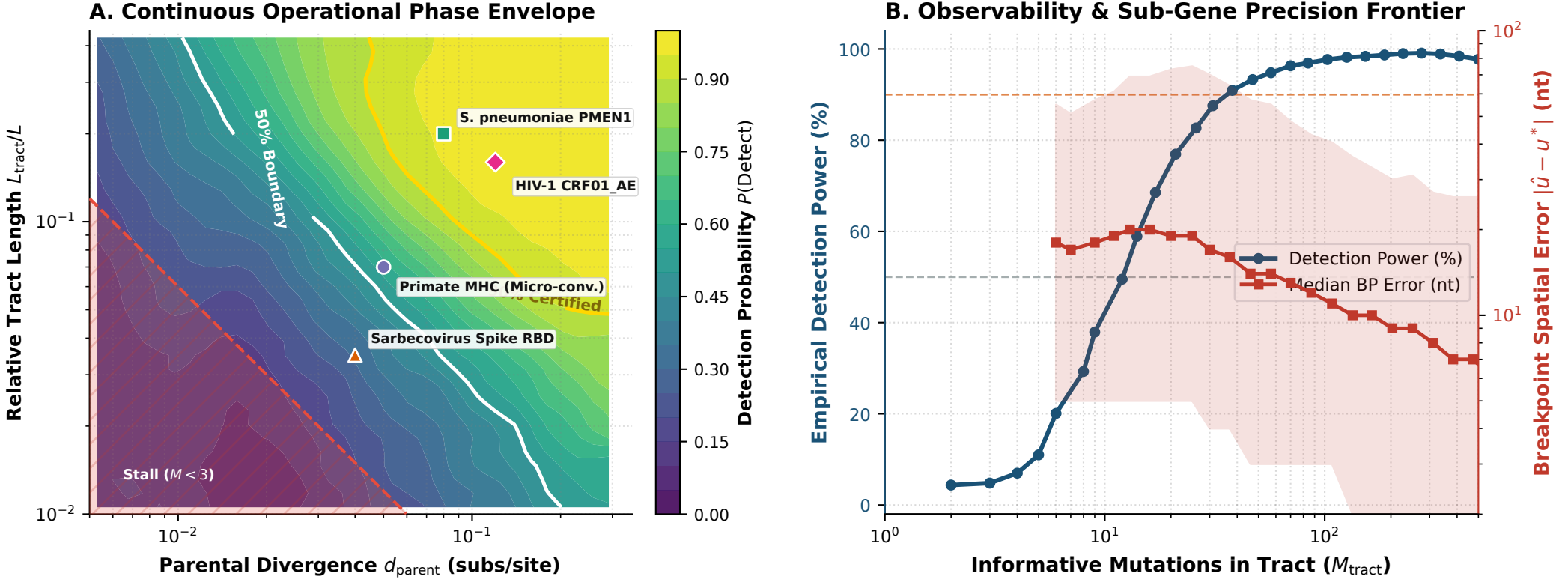
A. Continuous Operational Phase Envelope
Relative Tract Length L_{tract}/L
Parental Divergence d_{parent} (subs/site)
Detection Probability P(Detect)
50% Boundary
S. pneumoniae PMEN1
HIV-1 CRF01_AE
Primate MHC (Micro-conv.)
Sarbecovirus Spike RBD
Stall ($M < 3$)
B. Observability & Sub-Gene Precision Frontier
Empirical Detection Power (%)
Breakpoint Spatial Error $|\hat{u} - u^*|$ (nt)
Informative Mutations in Tract (M_{tract})
Detection Power (%)
Median BP Error (nt)